October 2026

Dr. Rishabh P. Sharma

Faculty of Sciences
Université libre de Bruxelles
Brussels, Belgium
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RESEARCH INTERESTS

Geological pattern formation, Reactive transport processes, porous media flows, geological soft matter

EDUCATION

PhD in Physics
Faculty of Physics, University of Warsaw, Poland — 2020 – 2025
Advisors: Prof. Piotr Szymczak and Prof. Mariusz Białecki
Thesis: Self-organization of flow in dissolving rocks

M.E. in Thermal Engineering (CGPA: 8/10)
Thapar University, India — 2015 – 2017
Advisor: Prof. Neeraj Kumar
Thesis: Nodal integral method for convection-diffusion transport in complex domains using linear and higher-order quadrilateral elements

B.Tech. in Mechanical Engineering (71.3%)
Uttar Pradesh Technical University, India — 2010 – 2014

RESEARCH EXPERIENCE

FRS-FNRS Postdoctoral Fellow
Faculty of Sciences, Université libre de Bruxelles, Belgium — Oct. 2025 – present
Advisor: Prof. Anne De Wit

Research Assistant
Institute of Theoretical Physics, University of Warsaw — Dec. 2018 – Nov. 2021
Advisors: Prof. Piotr Szymczak

Project Technical Assistant
Department of Energy Sciences and Engineering, IIT Bombay — Feb. 2018 – Nov. 2018
Advisors: Prof. Atul Kumar and Prof. Suneet Singh

Project Research Associate
Department of Energy Sciences and Engineering, IIT Bombay — Sep. 2017 – Feb. 2018
Advisors: Prof. Manaswita Bose

AWARDS, GRANTS AND FUNDING

- **FRS–FNRS Belgium mobility and congress grant** to attend the Belgrade School of Complex Systems, Belgrade, Serbia — 2026
- **Postdoctoral fellowship** awarded by Université libre de Bruxelles, Belgium — Oct. 2025 – Oct. 2027
- **Travel support** from Université Paris Cité to attend the Colloquium "Branching Networks", Paris, France — 2025
- **Research fellowship for CEUS-UNISONO project** funded by the National Science Centre (NCN), Poland, and the Slovenian Research Agency — Mar. 2022 – Feb. 2025

- **IDUB Rector Scholarship** awarded to top five theses of the year, University of Warsaw, Poland — Dec. 2024
- **Doctoral Scholarship** awarded by the Interdisciplinary Doctoral School (MSD), University of Warsaw, Poland — Oct. 2020 – Sep. 2024
- **IDUB research microgrant**, University of Warsaw, to attend the Geilo School, Norway — 2023
- **Early Career Scientist's Travel Support**, European Geosciences Union (EGU), Austria — 2023
- **Travel support** from the Integrated Development Programme (ZIP), University of Warsaw, for a research visit to the University of Minnesota, USA — 2023
- **IDUB research microgrant**, University of Warsaw, to attend the Geilo School, Norway — 2022
- SEG Student Chapter Field Camp 2020, Institute of Geophysics PAN, Poland — 2020
- **Research fellowship for OPUS-11 project** funded by the National Science Centre (NCN), Poland — Dec 2018 – Nov. 2021
- **Graduate Scholarship (GATE)**, Ministry of Education, Government of India — 2015
- **University Scholarship** for meritorious students — 2010
- **Bronze Medal**, Diploma in Programming, NCVT, Government of India — 2010

JOURNAL ARTICLES/ CONFERENCE PROCEEDINGS

1. **Sharma, R. P.**, Schuszter, G., and De Wit, A. (2026). Experimental investigation of SrCO3 phyllotactic precipitation patterns in Hele-Shaw cell. [In prep].
2. **Sharma, R. P.**, Cooper, M., and Szymczak, P. (2026). Geometric Characterization of Wormhole Shape in Core Dissolution and the Influence of Rock Structures. ***Advances in Water Resources***. [Ready to be submitted].
3. Deng, J., **Sharma, R. P.**, Szymczak, P., and Kang, P. K. (2025). Anomalous Transport in Dissolving Porous Media: Transitions Between Fickian and Non-Fickian Regimes. ***Geophysical Research Letters*** **52**(15), e2025GL115940.
4. **Sharma, R. P.**, Deng, J., Kang, P. K., and Szymczak, P. (2023). Effects of Mixing at Pore Intersections on Large-Scale Dissolution Patterns and Solute Transport. ***Geophysical Research Letters*** **50**(21), e2023GL105183.
5. **Sharma, R. P.**, Białecki, M., Cooper, M. P., Radliński, A. P., and Szymczak, P. (2023). Pore merging and flow focusing: Comparative study of undissolved and karstified limestone based on microtomography. ***Chemical Geology*** **627**, 121397.
6. Cooper, M. P., **Sharma, R. P.**, Magni, S., Blach, T. P., Radliński, A. P., Drabik, K., Tengattini, A., and Szymczak, P. (2023). 4D tomography reveals a complex relationship between wormhole advancement and permeability variation in dissolving rocks. ***Advances in Water Resources*** **175**, 104407.
7. **Sharma, R. P.** and Kumar, N. (2018). Nodal integral method for convection-diffusion transport using linear and higher order quadrilateral elements. ***Numerical Heat Transfer, Part B: Fundamentals*** **74**(3), 623–645.
8. **Sharma, R. P.** and Kumar, N. (2018). Nodal integral method for complex geometries using higher order elements. ***Proceedings of the 24th National and 2nd International ISHMT-ASTFE Heat and Mass Transfer Conference (IHMTC-2017)***. Begell House.

CONFERENCE PRESENTATIONS

1. **Sharma, R. P.**, Cooper, M. P., and Szymczak, P. (2025). Influence of rock structure on the morphology of wormhole networks. ***Colloquium "Branching Networks: From Experiment to Theory"***, 26–27 June 2025, Université Paris Cité, Paris, France.
2. **Sharma, R. P.**, Kang, P. K., and Szymczak, P. (2023). Effects of mixing at pore intersections on large-scale dissolution patterns. ***EGU General Assembly 2023***, 23–28 April 2023, Vienna, Austria.
3. **Sharma, R. P.** and Szymczak, P. (2021). Network modelling of the wormholing process in rocks. ***DREAMS21 Workshop***, 8–9 December 2021, Laboratoire Matière et Systèmes Complexes, Université Paris Cité, Paris, France.
4. **Sharma, R. P.**, Cooper, M. P., Ladd, A. J. C., and Szymczak, P. (2020). Flowfield study in dissolved porous media. ***CFD Wrocław-7 Workshop***, 18 June 2020, Institute of Theoretical Physics, University of Wrocław, Online.

CONFERENCE POSTERS

1. **Sharma, R. P.**, Schuszter, G., and De Wit, A. (2026). Experimental phyllotactic precipitation patterns. ***Belgrade School of Complex Systems***, 21–27 June 2026, Petnica, Serbia.
2. **Sharma, R. P.**, Cooper, M. P., and Szymczak, P. (2025). Influence of rock structure on the morphology of wormholes. ***8th Warsaw School of Statistical Physics***, 29 June–5 July 2025, Chęciny, Poland.
3. **Sharma, R. P.**, Deng, J., Kang, P. K., and Szymczak, P. (2024). Effects of mixing at pore intersections on large-scale dissolution patterns and solute transport. ***6th Cargèse Summer School***: Flow and Transport in Porous and Fractured Media (FLOWTIME), 10–21 June 2024, Cargèse, France.
4. **Sharma, R. P.**, Cooper, M. P., and Szymczak, P. (2023). Geometric measures of wormholes in dissolving rocks. ***Geilo School 2023***: The Physics of Evolving Matter Continued: Connectivity, Communication and Growth, 13–23 March 2023, Geilo, Norway.
5. **Sharma, R. P.**, Białecki, M., and Szymczak, P. (2022). How does chemical erosion change the pore structure of a rock? ***Geilo School 2022***: The Physics of Evolving Matter: Memory, Learning and Evolution, 21–31 March 2022, Geilo, Norway.
6. **Sharma, R. P.**, Kang, P. K., and Szymczak, P. (2022). The impact of intersection mixing rules on network-scale dissolution patterns. ***7th Warsaw School of Statistical Physics***, 26 June–2 July 2022, Sandomierz, Poland.
7. **Sharma, R. P.**, Cooper, M. P., Ladd, A. J. C., and Szymczak, P. (2021). Subpixel determination of wormhole tip position in 4D tomography of dissolving limestone cores. ***EGU General Assembly 2021***, 19–30 April 2021, Online.
8. **Sharma, R. P.**, Cooper, M. P., Ladd, A. J. C., and Szymczak, P. (2020). Modeling wormhole formation in digital rock samples: The role of segmentation and permeability–porosity relationships. ***EGU General Assembly 2020***, 4–8 May 2020, Online.

INVITED TALKS/ SEMINARS

1. **Sharma, R. P.**, Cooper, M. P., and Szymczak, P. (2021). Study of dissolution channels and their geometrical properties using XCMT images. ***Looping Network Meeting***, Université Paris Diderot, Paris, France.
2. **Sharma, R. P.** (2023). Physics of micro-scale cave formation in dissolving rocks. ***Interdisciplinary Doctoral School Seminar (MSD 2023)***, 27–28 May 2023, Chęciny, Poland.
3. **Sharma, R. P.** and Szymczak, P. (2023). Effects of mixing at pore intersections on large-scale dissolution patterns. ***Department of Earth and Environmental Sciences***, University of Minnesota, Minneapolis, MN, USA.

CONTRIBUTED POSTER/ PRESENTATIONS

1. Białecki, M., **Sharma, R. P.**, Cooper, M. P., and Szymczak, P. (2022). Comparative study of undissolved and karstified limestone based on microtomography. ***EGU General Assembly 2022***, 23–27 May 2022, Vienna, Austria. **[Talk]**.
2. Lipar, M., Szymczak, P., Ciglič, R., **Sharma, R. P.**, Zorn, M., Stepišnik, U., and Ferk, M. (2022). Challenges in characterisation and mapping of solution pipes. ***EGU General Assembly 2022***, 23–27 May 2022, Vienna, Austria. **[Talk]**.
3. Dzikowski, M., Szymczak, P., and **Sharma, R. P.** (2023). Empowering pre-exascale computers for Darcy–Brinkman simulation of wormhole growth based on X-CT data: Can we recover experiments? ***EGU General Assembly 2023***, 23–28 April 2023, Vienna, Austria. **[Talk]**.
4. Szymczak, P., Cooper, M. P., Magni, S., **Sharma, R. P.**, Blach, T. P., Radliński, A. P., Dohnalik, M., and Tengattini, A. (2020). Combined neutron and X-ray time-resolved tomography of wormhole growth in dissolving limestones. ***AGU Fall Meeting 2020***, 1–17 December 2020, Online. **[Talk]**.
5. Szymczak, P., Cooper, M. P., Magni, S., **Sharma, R. P.**, Blach, T. P., Radliński, A. P., Dohnalik, M., and Tengattini, A. (2021). Wormhole growth in dissolving limestones: Insights from 4D tomography. ***EGU General Assembly 2021***, 19–30 April 2021, Online. **[Talk]**.
6. Deng, J., Kang, P. K., **Sharma, R. P.**, and Szymczak, P. (2025). Anomalous transport in dissolving porous media: Transitions between Fickian and non-Fickian regimes. ***InterPore 2025***, 19–22 May 2025, Albuquerque, NM, USA. **[Poster]**.
7. Dzikowski, M., **Sharma, R. P.**, and Szymczak, P. (2023). High-resolution Darcy–Brinkman simulation of wormhole growth based on X-CT data. ***InterPore 2023***, 22–25 May 2023, Edinburgh, Scotland. **[Poster]**.
8. Cooper, M. P., Magni, S., **Sharma, R. P.**, Vu, P. N. H., Blach, T. P., Radliński, A. P., Dohnalik, M., Tengattini, A., Ladd, A. J. C., and Szymczak, P. (2019). Determining the influence of pore-scale geometry on wormhole formation. ***AGU Fall Meeting 2019***, 9–13 December 2019, San Francisco, CA, USA. **[Poster]**.
9. Deng, J., **Sharma, R. P.**, Szymczak, P., and Kang, P. K. (2023). Anomalous transport through dissolving fracture networks. ***AGU Fall Meeting 2023***, 11–15 December 2023, San Francisco, CA, USA. **[Talk]**.
10. Cooper, M. P., **Sharma, R. P.**, Magni, S., Vu, P. N. H., Ladd, A. J. C., Radliński, A. P., Blach, T. P., Drabik, K., Tengattini, A., and Szymczak, P. (2022). Competition between wormholes in dissolving rocks captured with 4D tomography and numerical models. ***AGU Fall Meeting 2022***, 12–16 December 2022, Chicago, IL, USA. **[Poster]**.
11. Dzikowski, M., Szymczak, P., and **Sharma, R. P.** (2022). Large-scale simulations of wormhole growth in dissolving porous media using the lattice Boltzmann method. ***AGU Fall Meeting 2022***, 12–16 December 2022, Chicago, IL, USA. **[Poster]**.

TEACHING

Graduate Teaching Assistant (Uni. of Warsaw)

Hydrodynamics and Elasticity	Fall 2023/2024

Student Team Project (Uni. of Warsaw)

Introduction to OpenFOAM (introduced by me)	Spring 2023/2024
Geophysical Laboratory II	Spring 2021/22
Study of the hydrodynamic properties of a mixture of ground coffee and water	Spring 2021/2022

Graduate Teaching Assistant (Thapar Uni.)

Advanced Fluid Dyamics	Fall 2016/2017
Advanced Heat Transfer	Spring 2016/2017

REFERENCES

Prof. Anne De Wit
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Prof. Peter Kang
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Prof. Anthony Ladd
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